{
"creator" : "ari.neko@cvut.cz",
"timestamp" : "09-07-2026_17:33:00",
"variant" : 1,
"subjectName" : "mathematics",
"categories" : [ "Planimetry", "Stereometry", "Mathematical Logic" ],
"questions" : [ {
"question" : "What is the area of a circle with a diameter of 10?",
"answers" : [ "25 * pi", "100 * pi", "10 * pi", "50 * pi" ],
"correctIndex" : 0
}, {
"question" : "What is the sum of the interior angles of a regular hexagon?",
"answers" : [ "720 degrees", "900 degrees", "360 degrees", "540 degrees" ],
"correctIndex" : 0
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"question" : "What is the length of the hypotenuse of a right-angled triangle with sides 5 and 12?",
"answers" : [ "13", "14", "15", "17" ],
"correctIndex" : 0
}, {
"question" : "How many faces does a regular octahedron have?",
"answers" : [ "8", "6", "12", "20" ],
"correctIndex" : 0
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"question" : "How many vertices does a standard cube possess?",
"answers" : [ "6", "8", "16", "12" ],
"correctIndex" : 1
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"question" : "What is the total surface area formula of a cylinder?",
"answers" : [ "2 * pi * r * (r + h)", "pi * r^2 * h", "pi * r * s", "2 * pi * r * h" ],
"correctIndex" : 0
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"question" : "What is the volume of a cone with base radius r and height h?",
"answers" : [ "pi * r^2 * h", "(1/3) * pi * r^2 * h", "(4/3) * pi * r^2 * h", "2 * pi * r * h" ],
"correctIndex" : 1
}, {
"question" : "Which logical connective represents an 'exclusive or' (XOR)?",
"answers" : [ "Both inputs are true", "At least one input is true", "Both inputs are false", "Exactly one input is true" ],
"correctIndex" : 3
}, {
"question" : "What is a logical statement called if it evaluates to true under all possible truth value assignments?",

"answers" : [ "Contingency", "Contradiction", "Tautology", "Fallacy" ],
"correctIndex" : 2
}, {
"question" : "What is the negation of the statement 'All primes are odd'?",
"answers" : [ "There exists at least one prime that is not odd", "No primes are odd", "Some primes are odd", "All primes are even" ],
"correctIndex" : 0
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